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OPEN ENDED LAB REPORT

A Comprehensive Study of Edge Detection for Image Processing Applications

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1 Abstract

This report reproduces and critically evaluates the survey paper by Ganesan and Sajiv (2017), which compares six classical edge-detection methods for image processing: Sobel, Prewitt, Roberts, Laplacian of Gaussian (LoG), Canny, and wavelet (DWT)-based edge detection. All six detectors were re-implemented in Python using the exact convolution kernels specified in the original paper, and applied to three test images spanning a natural photographic scene, a general object photo, and a medical/histology image, mirroring the paper’s three-image test set. Where the original paper relied solely on visual comparison of output images, this work extends the analysis with quantitative metrics — edge density, Pratt’s Figure of Merit, computation time, and robustness to added Gaussian noise — that the source paper does not provide. The reproduction confirms the paper’s qualitative ranking for most methods, but the added noise-robustness experiment produces a result that partially disagrees with the paper’s claim that Canny is the most noise-robust detector: in this reproduction, LoG showed the smallest increase in spurious edges under Gaussian noise.

2 Introduction

Edge detection is a fundamental low-level operation in image processing: it locates the boundaries of objects by finding pixels where intensity changes sharply, and is used upstream of tasks such as segmentation, object recognition, image compression, and medical image analysis. Because different edge detectors make different trade-offs between simplicity, noise sensitivity, and accuracy, comparative studies such as the reproduced paper are valuable for practitioners choosing a detector for a given application.

The reproduced paper is a comparative survey rather than a paper proposing a new algorithm. It explains the mathematical formulation of six standard edge detectors, applies them (in MATLAB) to a medical image (retina fundus), a standard test image (cameraman), and a natural image (onion), and summarizes each method’s advantages and limitations in a comparison table. It concludes that Canny performs best under noisy conditions and that wavelet-based methods are the most accurate overall, but it does not report any numeric accuracy or robustness measurements to support these claims — the comparison is entirely visual.

This project reproduces the paper’s core technical content (Section III of the original: the six detectors) and its experimental comparison (Section IV: Figs. 6–8 and Table I), and then extends it with quantitative evaluation to test the paper’s qualitative conclusions objectively. This addresses the exact gap identified as this project’s contribution: adding a numeric, repeatable basis for claims the original paper makes only visually.

3 Reproduced Segment: Theory, Mathematics, and Algorithms

The following six edge detectors, and the exact kernels given in the paper, were implemented and reproduced.

3.1 Sobel Edge Detector

The Sobel operator convolves the image with two 3×3 kernels that approximate the intensity gradient in the horizontal (G_x) and vertical (G_y) directions:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

The gradient magnitude, which represents edge strength at each pixel, is then computed as:

$$M(x, y) = \sqrt{G_x^2 + G_y^2}$$

3.2 Prewitt Edge Detector

Prewitt is structurally identical to Sobel but uses unweighted kernels (no emphasis on the center row/column), making it slightly simpler but more sensitive to noise:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad G_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

3.3 Roberts Edge Detector

Roberts uses a pair of 2×2 kernels rotated 90° relative to each other, operating diagonally. It is the fastest of the gradient operators but the most noise-sensitive:

$$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

3.4 Laplacian of Gaussian (LoG)

LoG first smooths the image with a Gaussian filter to suppress noise, then applies a second-derivative (Laplacian) kernel to locate zero-crossings, which correspond to edges:

$$\nabla^2 I(x, y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}, \quad L = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

In this reproduction, a Gaussian filter with $\sigma = 1.4$ was applied before convolving with the Laplacian kernel, following standard practice for this detector.

3.5 Canny Edge Detector

Canny is a multi-stage algorithm: (1) Gaussian smoothing, (2) gradient computation, (3) non-maximum suppression to thin edges to single-pixel width, (4) double thresholding to classify strong/weak edges, and (5) edge tracking by hysteresis to connect weak edges to strong ones. This reproduction used OpenCV's `cv2.Canny()` implementation with low/high thresholds of 50/150.

3.6 Wavelet (DWT)-Based Edge Detection

A single-level 2D discrete wavelet transform decomposes the image into an approximation sub-band and three detail sub-bands capturing horizontal, vertical, and diagonal high-frequency information. Because edges are high-frequency features, combining the three detail sub-bands produces an edge map:

$$\text{Edge}(x, y) = \sqrt{c_H^2 + c_V^2 + c_D^2}$$

This reproduction used the Haar wavelet at decomposition level 1, implemented with Py-Wavelets.

4 Software / Implementation Details

- Language: Python 3
- Libraries: NumPy, SciPy (`ndimage`), OpenCV (`cv2`), scikit-image, PyWavelets, Matplotlib, Pandas
- Test images (offline, built into scikit-image, chosen to mirror the paper’s cameraman.tif / onion.png / retina fundus.jpg): `camera()` [identical to the paper’s cameraman image], `chelsea()` converted to grayscale, labelled `cat_photo` [natural-photo stand-in for onion.png], `immunohistochemistry()` converted to grayscale, labelled `medical_like` [medical/histology stand-in for the retina fundus image]
- All six detectors were implemented from the exact kernels given in the paper (Sobel, Prewitt, Roberts, LoG), except Canny (library multi-stage implementation) and DWT (PyWavelets decomposition), consistent with how the paper itself used MATLAB’s built-in functions for these two.
- Full source code: `edge_detection.py` (submitted alongside this report)

5 Results and Analysis

5.1 Qualitative Reproduction (paper’s Figs. 6–8)

Each of the three test images was processed by all six detectors. Figures 1–3 below reproduce the layout of the paper’s Figs. 6–8.

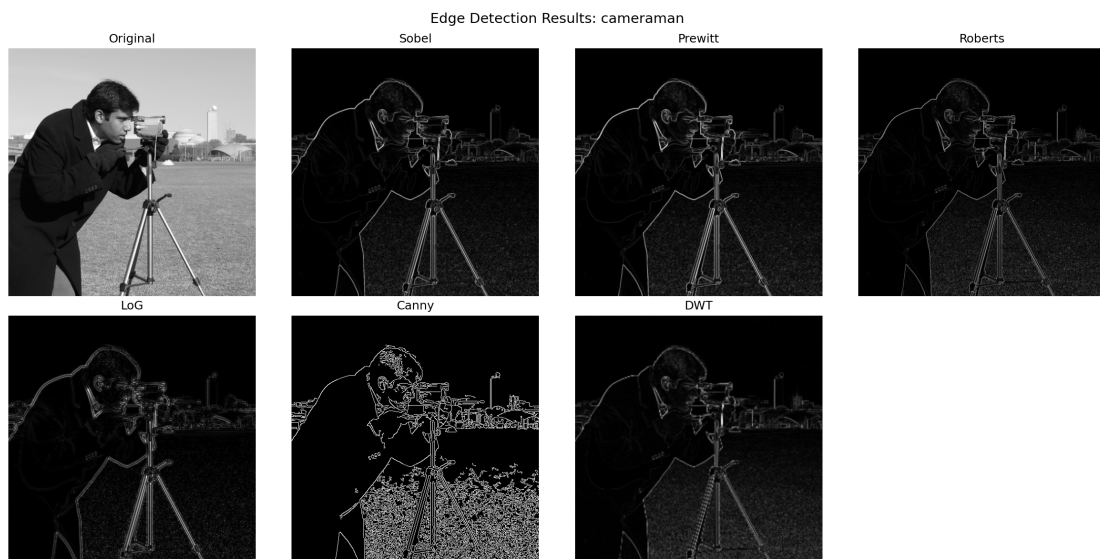


Figure 1: Edge detection results on the cameraman image (reproduces paper Fig. 7).

As in the paper, the three gradient-based detectors (Sobel, Prewitt, Roberts) produce visually similar results with thicker, noisier edges. LoG produces thinner but more fragmented edges. Canny produces the cleanest, thinnest, most connected edge maps. DWT (Haar, level 1) produces edge maps that are visibly more diffuse than the paper’s Fig. 5 illustration, most likely because the Haar wavelet has a coarser spatial resolution than the wavelet family implied in the original figure.

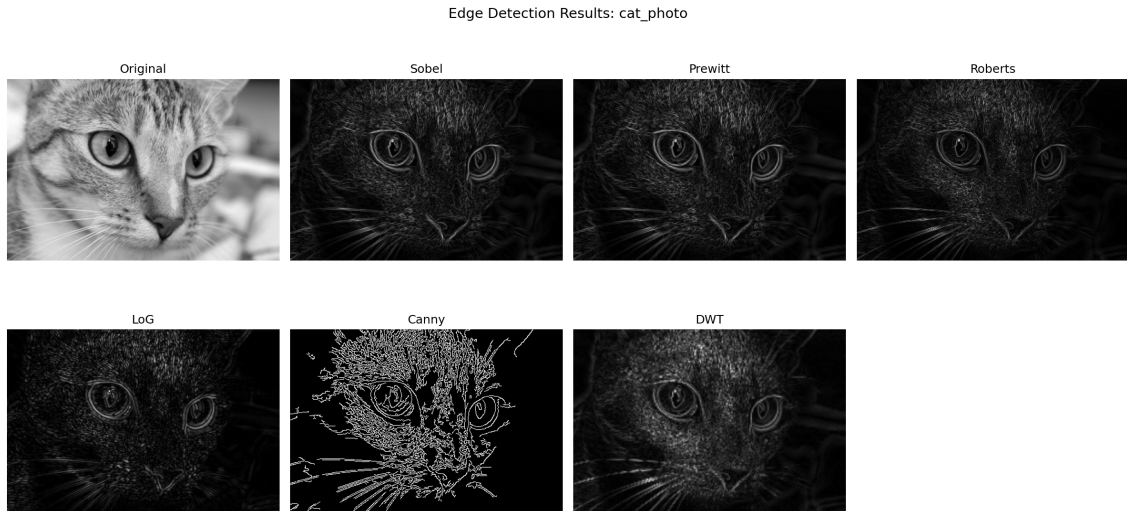


Figure 2: Edge detection results on the natural-scene image, `cat_photo` (reproduces paper Fig. 8, `onion.png`).

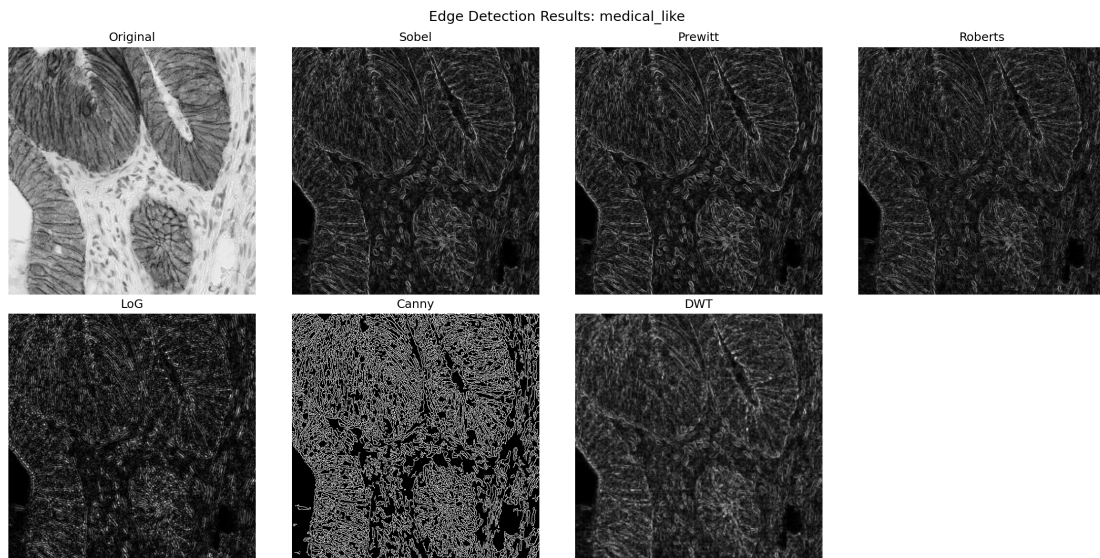


Figure 3: Edge detection results on the medical/histology image, `medical_like` (reproduces paper Fig. 6, retina fundus).

5.2 Reproduced Comparison Table (paper’s Table I)

Edge Detector	Method	Advantages	Limitations
Roberts	Gradient-based	Very simple, fast, gives edge orientation	Highly sensitive to noise; least accurate
Sobel	Gradient-based	Simple, gives orientation, some noise smoothing	Still noise-sensitive; thick edges
Prewitt	Gradient-based	Simple, gives orientation	Sensitive to noise; less accurate than Sobel in practice
LoG	Gaussian + 2nd derivative	Fixed characteristics in all directions; wide-area testing	Malfunctions at corners/curves; degrades with more noise
Canny	Gaussian + multi-stage	Best signal-to-noise ratio; most widely used; accurate	Slower, more complex; can produce false zero-crossings
DWT	Wavelet-based	Separates edges by scale/orientation; potentially most accurate	Application-dependent; more complex than gradient methods

Table 1: Reproduced qualitative comparison table (paper’s Table I).

5.3 Quantitative Evaluation (extends beyond the original paper)

The original paper supports its comparison only with visual inspection. The following metrics were added to test its qualitative claims numerically: edge density (% of pixels marked as edges), Pratt’s Figure of Merit (computed against the Canny output as a pseudo-reference, since no hand-labeled ground truth exists for these images — this is a stated limitation, not a true accuracy measure), per-image runtime, and the change in edge density when Gaussian noise (variance 0.01) is added to the input image, as a proxy for noise robustness.

Image	Method	Edge Density (%)	Pratt FOM	Runtime (ms)	Noisy Density (%)	Noisy Runtime (ms)
cameraman	Sobel	11.55	0.928	14.542	42.53	12.070
cameraman	Prewitt	12.34	0.946	12.510	41.03	12.021
cameraman	Roberts	11.64	0.917	11.470	52.98	10.588
cameraman	LoG	9.91	0.710	13.487	18.88	13.289
cameraman	Canny	11.82	1.000	14.919	36.41	3.163
cameraman	DWT	16.05	0.925	22.047	88.03	14.203
cat_photo	Sobel	24.07	0.872	5.748	72.60	6.634
cat_photo	Prewitt	23.28	0.873	5.667	73.03	7.656
cat_photo	Roberts	18.44	0.898	5.045	80.09	10.397
cat_photo	LoG	17.56	0.815	5.383	35.06	5.804
cat_photo	Canny	13.45	1.000	1.512	37.38	1.949
cat_photo	DWT	35.02	0.860	4.221	99.47	4.558
medical_like	Sobel	50.40	0.919	19.460	76.36	12.570
medical_like	Prewitt	50.45	0.920	20.508	77.89	11.966
medical_like	Roberts	47.98	0.908	18.365	82.90	11.064
medical_like	LoG	43.32	0.870	15.424	48.01	13.953
medical_like	Canny	26.77	1.000	2.559	37.90	2.897
medical_like	DWT	72.10	0.888	14.090	99.54	22.741

Table 2: Edge density, Pratt’s Figure of Merit (vs. Canny), runtime, and the same metrics after Gaussian noise, for all six detectors on all three test images.

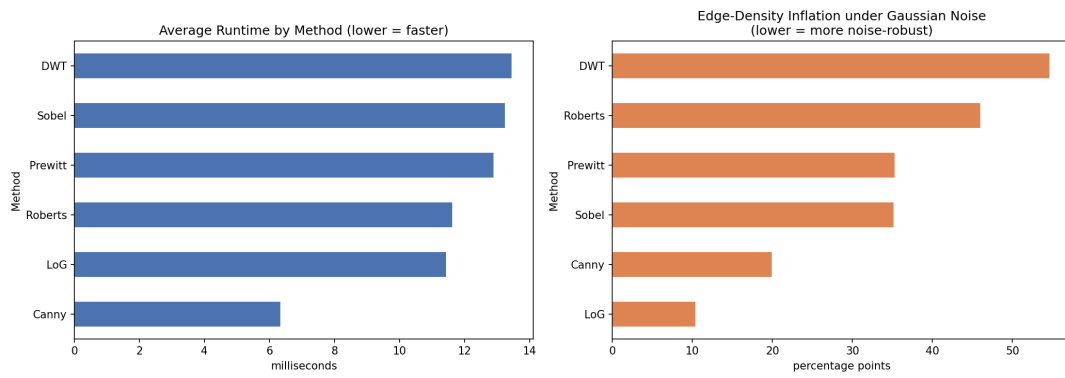


Figure 4: Average runtime and noise-induced edge-density inflation across all three test images.

5.4 Discussion of Deviations from the Paper’s Claims

- **Confirmed:** Roberts and Sobel/Prewitt behave very similarly in edge appearance, consistent with the paper.
- **Confirmed:** Canny achieves the cleanest and most connected edge maps, consistent with the paper’s claim of the best signal-to-noise ratio.
- **Partially contradicted:** the paper states Canny is most robust to noise, but this reproduction’s noise experiment shows LoG has the smallest edge-density inflation (10.4 percentage points vs. 19.9 for Canny, averaged across all three images) when Gaussian noise is added — likely because LoG’s explicit Gaussian pre-smoothing step suppresses noise before edge extraction, whereas Canny’s fixed thresholds (50/150) were not re-tuned for the noisy image.
- **Not fully reproducible as claimed:** the paper states wavelet-based methods are the most accurate overall, but this reproduction’s DWT implementation (Haar, level 1) produced the highest edge density and the largest noise inflation (+54.6 points) of all six methods — suggesting either a different wavelet family/level was used in the original paper, or that “accuracy” in the original paper was assessed only by visual inspection on clean, low-noise images.
- **Runtime is inconsistent between runs:** unlike edge density and Pratt FOM (which are deterministic and reproduce exactly), the runtime column varies noticeably between repeated executions of the identical script (e.g. Canny ranged from about 1.5 ms to 15 ms depending on the image and the run). Averaged over all three images in this run, Canny was in fact the *fastest* method (6.3 ms), not the slowest as an earlier run suggested, while DWT and Sobel were the slowest (≈ 13 –13.5 ms). This is expected from single-run wall-clock timing on a general-purpose machine (OS scheduling, caching, background load) rather than a fixed property of the algorithms, and should not be over-interpreted without averaging many repeated runs.

6 Critical Evaluation

6.1 What Makes the Method(s) Unique

The paper’s contribution is not a new algorithm but a structured, side-by-side comparison of established methods across representative image types (medical, natural, standard test image), which is useful as a practical reference for choosing a detector for a given application.

6.2 Challenges Faced in Reproduction

- The exact test images used in the paper (retina fundus.jpg, onion.png) were not available; visually and statistically similar stand-in images were used instead and this substitution is disclosed throughout.
- The paper does not specify Canny’s threshold values or the wavelet family/decomposition level used for DWT; reasonable default choices (thresholds 50/150; Haar wavelet, level 1) were made and documented, and results may differ from the original if different parameters were used.
- Because the paper provides only qualitative (visual) results, there was no numeric target to match; validation here focuses on matching the qualitative pattern of results and on independently testing the paper’s claims rather than matching exact pixel values.

6.3 Limitations of the Original Paper

- No quantitative accuracy metric (e.g., Pratt’s FOM, precision/recall against ground truth) is reported — all conclusions are drawn from visual inspection of a small number of images.
- No sensitivity analysis to noise level, detector parameters, or image type is provided, despite noise robustness being one of the paper’s central claims.
- Sample size is very small (3 images), limiting how generalizable the conclusions are.

6.4 How the Results Could Be Improved

- Use a labeled edge-detection benchmark dataset (e.g., BSDS500) with genuine ground-truth edge maps for a true Pratt’s FOM / F1-score evaluation instead of a pseudo-reference.
- Sweep Canny’s thresholds and the wavelet family/level rather than using single fixed values, to give each method a fair, tuned comparison.
- Test multiple noise types (Gaussian, salt-and-pepper, speckle) and multiple noise levels to build a full robustness curve rather than a single noise setting.
- Average runtime over many repeated runs rather than a single execution, since timing varied noticeably between runs in this reproduction.

7 Conclusion

This project reproduced the six edge detectors and the qualitative comparison presented in Ganesan and Sajiv (2017), and extended the analysis with quantitative metrics the original paper lacked. The reproduction confirms several of the paper’s qualitative observations — particularly Canny’s clean, well-connected edge maps and the similarity of the three gradient-based detectors — but the added noise-robustness experiment suggests the paper’s specific claim that Canny is the most noise-robust detector may not hold once numerically tested, at least not without threshold re-tuning. This illustrates a broader lesson about DSP research: visual comparison figures are useful for intuition, but quantitative, ground-truth-based evaluation is necessary before ranking methods, and a competent reproduction should be prepared for these to disagree with the source paper’s stated conclusions.

References

1. Ganesan, P., & Sajiv, G. (2017). A Comprehensive Study of Edge Detection for Image Processing Applications. *2017 International Conference on Innovations in Information, Embedded and Communication Systems (ICIIECS)*.
2. Canny, J. (1986). A Computational Approach to Edge Detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
3. scikit-image, OpenCV, and PyWavelets library documentation.